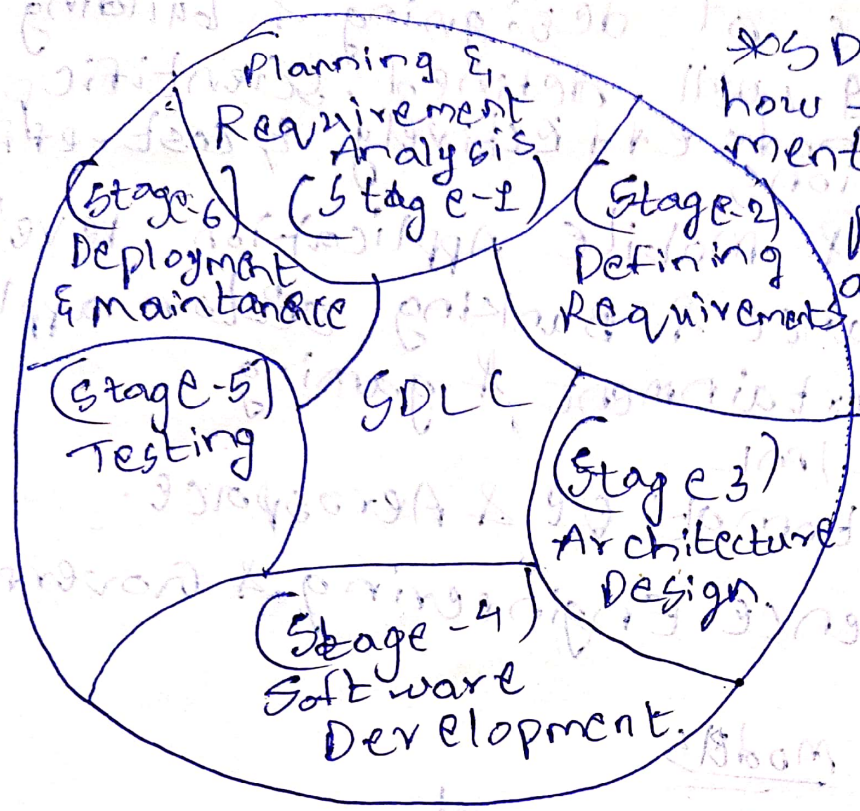
SDLC Diagram (Software Development Life Cycle)



*SDLC narrates how the development process progress as the Software Development takes place.

25/07/26

Waterfall model

* Waterfall model is a linear and sequential SDLC model in which each phase must be completed before the next phase begins.

→ Sequential flow of activities
→ Documentation-driven process
→ Clear milestones & deliverables.

Characterstics of software

→ Functionality
→ Efficiency
→ Reliability
→ Maintainability
→ Portability
→ Usability.

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Software Engineering.

* Process of designing & building product using well defined, scientific principles & methods effectively & cost-effectively.

Applications

→ Web & Mobile Application development.
→ Finance & Banking sector applications.
→ Entertainment & gaming.
→ AI/ML
→ Automative & Aerospace.
→ Science Engineering & Government.

Process Models

→ Waterfall Process model
→ V model
→ Incremental model
→ Iterative model
→ Spiral model
→ Prototype model
→ RAD model (Rapid Application Development).
→ Big Bang model
→ Component-Based model.
→ Unified process model
→ Rational Unified Process (RUP) } model
→ Personal Software Process (PSP)
→ Team Software Process (TSP)
→ Secure Software development Lifecycle.

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Adaptive Process models

→ Agile (framework) model.
* Scrum, KANBAN, XP Programming, SAFE
→ DEVOPS model

Agility

→ Ability to move quickly, easily & efficiently also adaptable to changing situations.
* Improves product quality.

Agile Manifesto

1) Individuals & Interactions over processes & tools.
2) Working software over comprehensive documentation.
3) Customer Collaboration over contract negotiation.
4) Responding to change over a following plan.

01/08/26

List of Agile Methodoligies

→ Scrum
→ kanban
→ Extreme Programming (XP)
→ Lean Software Development
→ Feature Driven Development (FDD).
→ Dynamic Systems Development Method (DSDM)
→ Crystal Methodoligies.
→ Agile Unified Process (AUP).

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→ Disciplined Agile Delivery (DAD).
→ Scaled Agile Framework (SAFe).
→ Rapid Application Development (RAD)
→ Adaptive Software Development (ASD).
→ Scrumban
→ Enterprise Scrum.
→ Spotify Model
→ DevOps.

SCRUM FRAMEWORK

* SCRUM is an IID (Iterative Incremental Development method.
* This emphasizes a set of project management values & practices, rather than those in requirements, implementation & so on.
* Used in complex projects, especially software development.

Key Practices

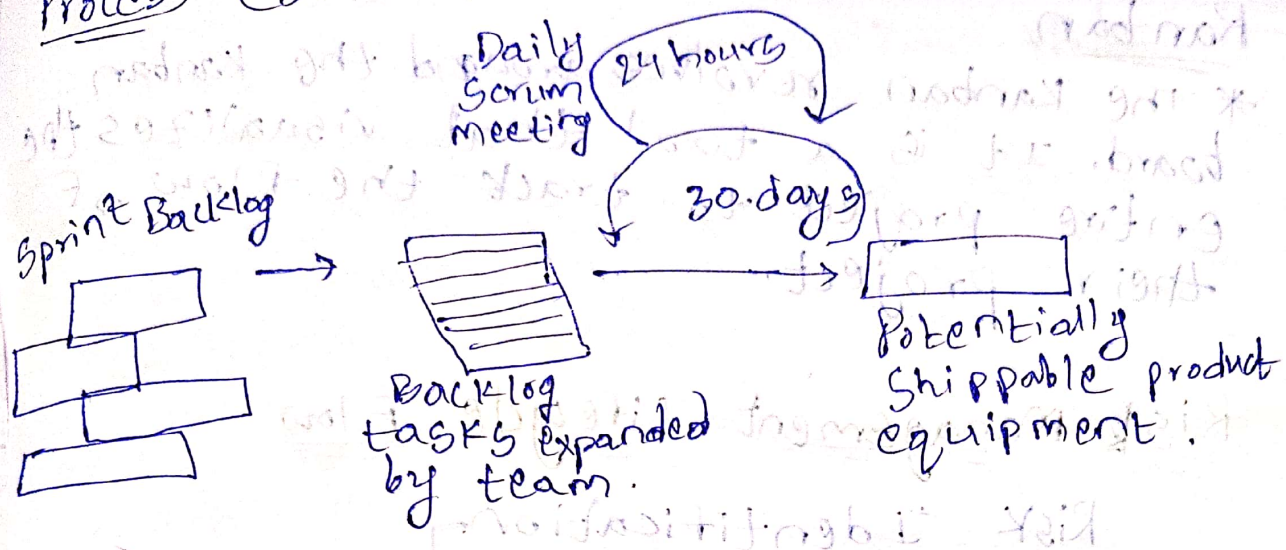
→ Self-directed & self-organizing teams.
→ No external addition of work to an iteration.

Characteristics

→ Agile & lightweight process.
→ Can manage & control software development.
→ Increases productivity.

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Process (Scrum).



SCRUM Frame Work

Roles

→ Product Owner
→ Scrum Master
→ Team.

Ceremonies

→ Sprint Planning.
→ Sprint Review
→ Sprint Retrospective
→ Daily Scrum meeting.

Artifacts

→ Product backlog. = list of requirements.
→ Sprint backlog = tasks selected from the product Backlog.
→ Increment = Working product delivered.

SCRUM Benefits

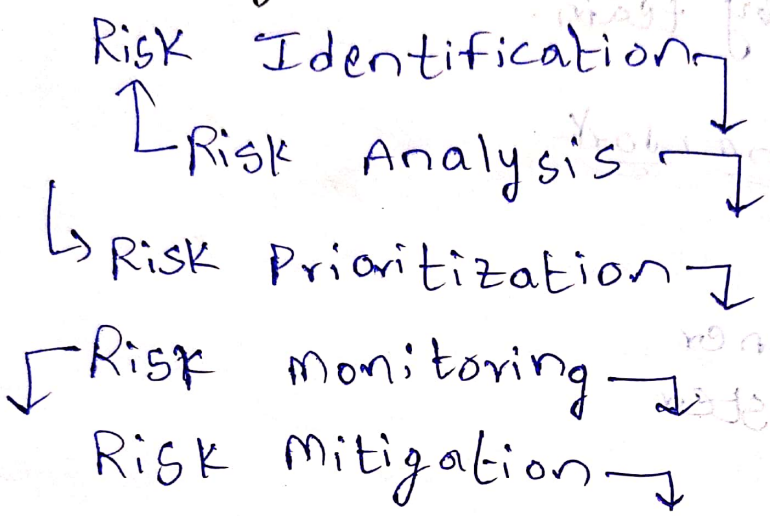
→ Faster Product Delivery.
→ Adaptability to Changes.
→ Improved Team Collaboration.
→ Improvement.

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Kanban

* The Kanban revolves around the kanban board. It is a tool that visualizes the entire project to track the flow of their project.

Risk Management Lifecycle Flow



Risk mitigation Strategies

1) Risk Avoidance
2) Risk Reduction
3) Risk Transfer
4) Risk Acceptance.

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14/08/26

DevOps (Development & Operations)

* It is an approach or culture that makes things better.
* It describes how it can be used to make development & operations teams efficient & effective in order.

→ Plan
→ Create
→ Verify
→ Package
} Development

→ Release
→ Configure
→ Monitor
} Operations

DevOps Lifecycle

- Develop
  ↓
  Test
  ↓
  Provision
  ↓
  Configure
  ↓
  Deploy
  ↓
  Monitor
  ↓
  Feedback

* DevOps is an agile relationship between Development & IT operations.

Principles

→ Customer-centric action
→ End-to-End responsibility
→ Automate everything you can
→ Continuous improvement
→ Cross-functional autonomous teams
→ Create with the end in mind.

* Devops is used to develop the organizations ability to deliver specifications & services at high velocity.

Characteristics

→ Culture
→ Automation
→ Lean
→ Measurement
→ Sharing.
→ Continuous integration & Delivery
→ Automated testing & deployment.

Docker for Containerization

29/08/

* A container is a lightweight & portable Software package that contains an application along with all the libraries, dependencies, runtime environments, & configuration files required for execution

Containerization

* With containerization, applications can run reliably on any platform supporting containers.

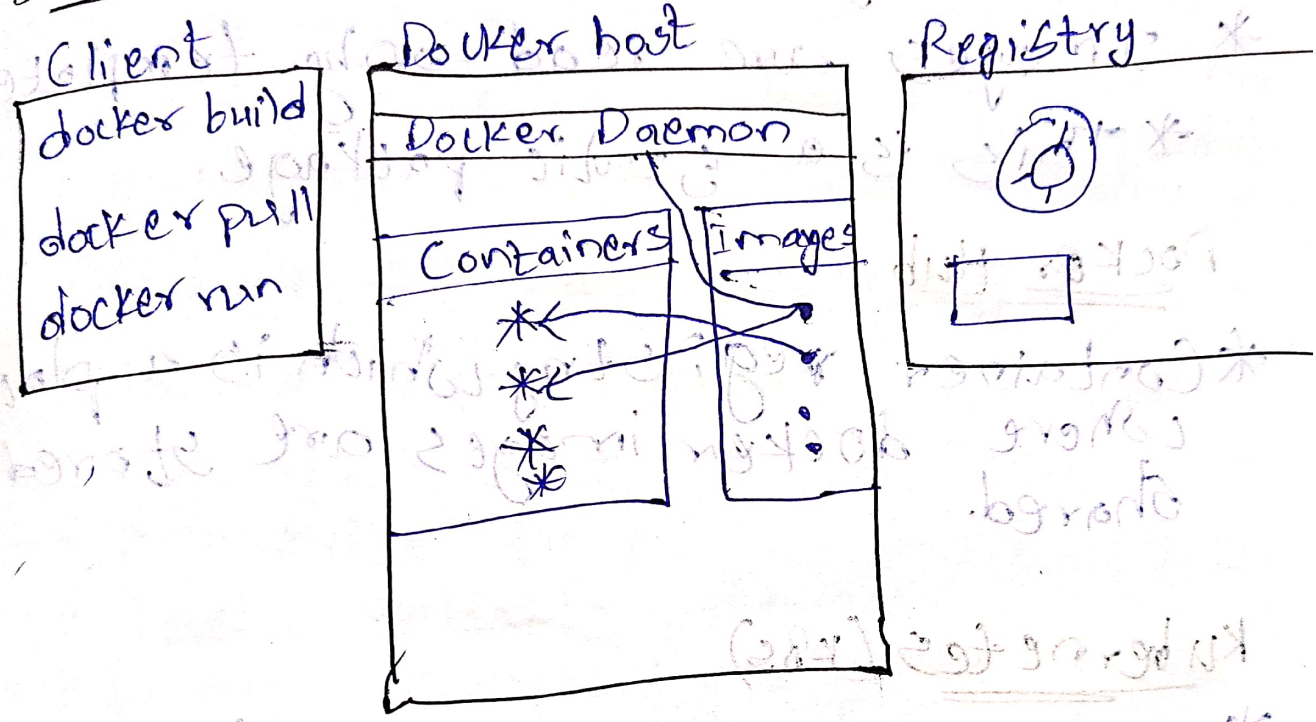
Source Code
↓
Dockerfile
↓
Docker Image
↓
Docker Registry
↓
Docker Container

* We will not have the guest operating system in the containerized environment

Docker

* Docker is an open platform that enables users to develop, ship & run applications with ease.

Docker Architecture



Docker Daemon

* This is the brain of the docker.
  - → Builds containers
  - → Runs containers
  - → Manages images
  - → Handles networks & volumes.

Docker client

* This is the tool used to talk in Docker.
* It is like a command Line Interface (CLI).

Docker Registries

* A registry is a storage place where Docker images are kept.
  * Default registry ⇒ Docker Hub.

Docker Images

* Images are read-only templates
  * This is a static package.

Docker Hub

* Container registry which is a place where docker images are stored & shared.

Kubernetes (K8s)

* This is an open-source container orchestration platform originally developed by Google.
* It automates the deployment, scaling, networking & management of containerized applications.
*

Software

* It is a program or set of programs containing instructions.

Characterstics

→ Functionality,
→ Efficiency
→ Reliability.

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→ Maintainability.
→ Portability
→ Usability.

Dual Role

* First one is Software is a Product.
* Second one is Software is used for developing & delivering another product

Need of Software Engineering

→ Handle Huge Projects
→ Manage cost
→ Minimize time
→ Develop reliable software
→ Reduce complexity
→ Improve Productivity & Efficiency.

Adoptive Software Engineering

*It is a flexible, Agile methodology designed to embrace changing requirements and uncertainity.

Software Engineering Process

*Set of instructions in the form of programs which govern the computer system & it's associated hardware to provide the desired functions or functionality.

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Software Engineering Process Models

*These process models are used to tailor the sequence of activities so as to develop the software efficiently.

The phases in Process Model

* Analysis
* Design
* Coding
* Testing
* Deployment
* Maintanence

SDLC Phases

1) Planning & Analysis

→ Stakeholders define the Project scope, feasibility analysis, cost estimation, scheduling, resource planning are done in this phase.

2) Requirement Gathering

→ Here analysts collect precise busine and user needs to create the Software Requirements Specification (SRS) document

3) System Design

→ Here requirements are transformed into a technical blueprint for implementation.

1) High-level Design ⇒ Defines system architecture, tech stack, database design & major modules.

2) Low-level Design ⇒ specifies component logic, API's, data structures & workflows.

4) Development -

*Software developers build the software based on the approved design.

5) Testing

*Ensures that software meets requirements & is free from errors.

Types

1) Unit Testing : Verifies individual components.

2) Integration Testing: Ensures modules work together.

3) System Testing: Validates the complete system.

4) User Acceptance Testing: Confirms business requirements are met or not.

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6) Deployment

* Tested software is released to users.

7) Maintainice

* Post-deployment support ensures long-term usability.

Waterfall Model

* This is a linear & sequential Software Development Life cycle model in which each phase must be completed before the next phase begins.

Characterstics

→ Sequential flow of activities
→ Documentation-driven process
→ Clear milestones & deliverables.
→ Minimal overlap blw phases.

Diagram

Requirements
↓
Design
↓
Development
↓
Testing
↓
Deployment → Maintanence

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Incremental Model

* This is an SDLC model in which the software is developed, tested, & delivered in small, manageable parts called increments.
* The project is divided into multiple releases.

features

→ Software is developed in small increments.
→ Each increment is independently tested.
→ Customer feed back is incorporated continuously.
→ Easier defect detection & correction.

Prototyping model

* In this a preliminary version of the Software, called prototype, is developed to understand & refine user requirements before building the system.

final

Key Idea ⇒ Builds a working model first then gathers the user feedback, refines the requirements & develops the final software.

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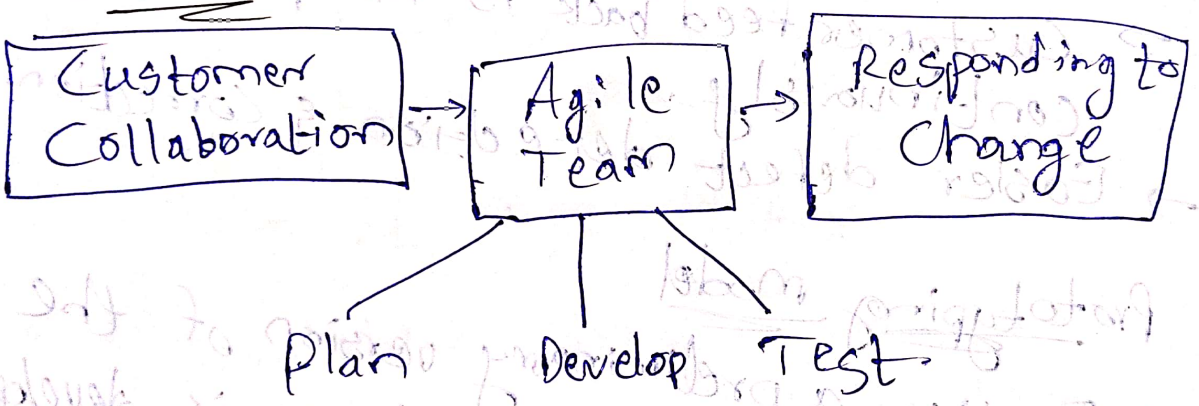
Agile Methodology

*Agility is the ability to move quickly, easily, and efficiently, while also being able to adapt to changing situations.

* Agile Model

→ This is a software development approach in which a project is developed in small, manageable parts called iterations or sprints.

Workflow



Agile Process steps

→ Requirements
→ Design
→ Code
→ Test
→ Release.

* After each cycle, feedback is collected and improvements are made for the next iteration.

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Benefits of Agile

→ Faster delivery of software
→ Improved product quality.
→ Better customer satisfaction.

Agile Lifecycle

→ Product Backlog = Collection of requirements.
→ Sprint Planning = Team selects plans & tasks.
→ Development = Developer build the selected features.
→ Testing
→ Review = Completed work is shown to customers & feedback is collected.
→ Retrospective = Team analyses the process & identifies improvements.
→ Release = Completed features are delivered.

Agile	Traditional
→ Development is done in small cycles	→ Development is done step-by-step in fixed phases.
→ Requirements can change during development	→ Requirements are fixed at the beginning.
→ Customer feedback taken continuously	→ Customer involvement is mostly at the beginning.
→ Suitable for projects with changing requirements.	→ Suitable for projects with clear & stable requirements.

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~~Scrum~~

User Stories

→ User Story Format

* As a ... (Role)
* I want ... (Goal)
* So that ... (Value/Benefit)

Estimation Techniques

→ Planning Poker

* Team members use cards (0, ½, 1, 2, 3, 5, 8 to estimate stories collaboratively.

→ t-shirt sizing

→ Wideband Delphi

→ Three point Estimation

Agile Estimation

* Agile Estimation is relative, not absolute.
* Estimate to understand, not to be perfect.

Estimation velocity

* Amount of work completed in a sprint.

No of story points completed = velocity

(or)

Sum of completed story points in a sprint

* Velocity helps decide how much work to take into a sprint.

Agile tools

* These are software tools that help teams plan, track, manage & deliver projects using Agile methodologies like scrum & kanban.

→ Enable real-time collaboration.
→ Tracks progress
→ Flexibility.
→ Faster delivery.

Popular Agile Tools

→ Jira = Used for Sprint planning, bug tracking & reporting.
→ Trello = Simple board-based tool for task management.
→ Asana = Project tracking with timelines & workflows.
→ Clickup = All-in-one productivity & Agile management tool.

Jira

* Developed by Atlassian.

→ Issue Tracking
→ Agile Boards.
→ Sprint Management
→ Backlog Management
→ Report's & Analytics
→ Customization

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Jira Boards => These are visual tools in Jira that help teams track & manage tasks during project execution.

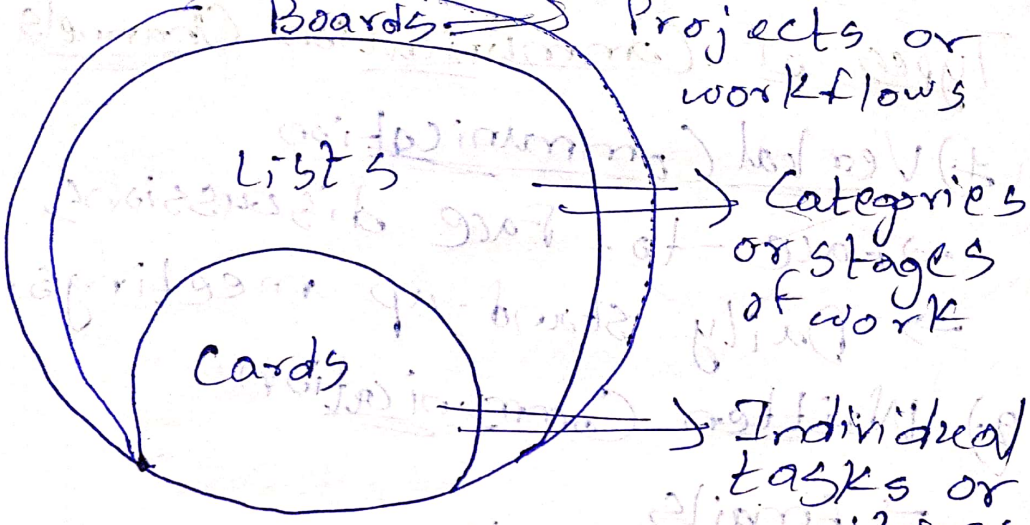
Types
→ Scrum Board
→ Kanban Board

Elements of Boards
→ Columns = Represent stages
→ Cards = Individual tasks (or) bugs.
→ Swimlanes = Group tasks by priority, User stories.
→ Backlog = List of pending tasks.

⇒ Workflow defines how work progress
⇒ Board visually displays that workflow

Trello
* Developed by Atlassian.
* Project management tool.
* Kanban - style board system.
→ Boards
→ Lists
→ Cards
→ Drag & Drop Interface
→ Collaboration
→ Labels & Tags.

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Team Communication in Agile

*Team Communication in Agile refers to the continuous, transparent exchange of information.

→ Enhances Collaboration.
→ Ensures Clarity of goals.
→ Faster Problem Resolution.
→ Improves Transparency.
→ Supports Continuous Feedback.
→ Increases Productivity.

Agile Team Collaboration

*This is the process where cross-functional team members work together continuously to deliver high-quality software in iterative cycles.

Stakeholder Communication

*Stakeholder communication is interaction between the Agile team and stakeholders to ensure that the product meets expectations.

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Types of Communication Channels in AGILE

1.) Verbal Communication
   → Face-to-Face discussions
   → Daily stand-up meetings.
2.) Written Communication
   → Emails
   → Documentation
   → Reports.
3.) Digital Communication
   → Chat tools (Slack, Microsoft Teams)
   → Project tools (Jira, Trello).
4.) Visual Communication
   → Kanban & Scrum boards.

Distributed Agile Teams

* In these teams the members work from different geographical locations but collaborate using agile principles

~~Contin~~